MODULE 5A · EPISODE 2

The Earth’s Atmosphere: Weather and Climate Systems

The Physics Behind the Weather That Surrounds Us

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

The Earth's Atmosphere: Weather and Climate Systems

This episode descends from the layers explored in Episode 1 into the troposphere, where all of Earth's weather happens. The panel treats weather as physics: thermodynamic events driven by heat moving from warmer places to cooler ones. It works through six ingredients in turn: the atmosphere itself; water, with its precipitation, hail, lightning, clouds, evaporation, and humidity; the Sun, with albedo and greenhouse gases; Earth's rotation, with convection cells, the Coriolis Effect, and global winds; volcanic activity; and biological activity, closing with how human activity is changing the climate.

The Panel

Name	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Guest; cosmologist, European Space Agency
Dr. Amara Johnson	Guest; theoretical physicist, atmospheric physics

Learning Objectives

After this episode you should be able to:

- Describe weather as thermodynamic events in the troposphere and name its six main ingredients.
- Explain how rain, snow, sleet, and hail form, and how charge separation in a cloud produces lightning and thunder.
- Use the flash-to-thunder delay to estimate the distance to a lightning strike.
- Explain how clouds form around condensation nuclei, why evaporation cools, and how latent heat feeds a cloud.
- Define relative humidity and dew point, and use them to explain dew and dry winter indoor air.
- Explain albedo and the greenhouse effect, and why nitrogen and oxygen are not greenhouse gases.
- Connect differential heating, convection cells, and the Coriolis Effect to global wind patterns.
- Define temperature and convert between the Fahrenheit and Celsius scales.
- Compare the short-term and long-term climate effects of volcanoes with the effects of human activity.

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QUICK REFERENCE

Key Terms

Term	Definition
Weather	Thermodynamic events in the troposphere, driven by the movement of heat due to temperature differences.
Thermodynamic event	An event driven by the movement of heat from warmer to cooler regions.
Meteorology	The study of weather; from the Greek *meteoros*, "lofty" or "high in the air."
Precipitation	Water falling from the sky as rain, snow, sleet, or hail.
Sleet	Small pellets of ice that form when falling raindrops freeze before reaching the ground.
Hail	Layered ice balls built up as updrafts carry ice particles through supercooled water droplets.
Graupel	Soft hail pellets; collisions between graupel and ice crystals separate electric charge in a cloud.
Lightning	A giant electrical discharge that releases the voltage difference built up in a storm cloud.
Thunder	The sound of the shock wave made when lightning heats the air so suddenly that it expands explosively.
Condensation nuclei	Tiny airborne particles (dust, smoke, sulfates, bacteria) around which cloud droplets and ice crystals form.
Evaporation	The persistent transfer of fast-moving water molecules from liquid water into the air above it; a cooling process.
Latent heat	Energy carried off by evaporating water and released again when the vapor condenses.
Relative humidity	Current water vapor pressure divided by the saturation vapor pressure, times 100%.
Saturation vapor pressure	The maximum water vapor pressure air can hold at a given temperature; rises with temperature.
Dew point	The temperature to which air must cool for its relative humidity to reach 100%.
Homosphere	The well-mixed lower atmosphere: troposphere, stratosphere, and mesosphere.
Albedo	Earth's reflectivity; the fraction of incoming sunlight reflected straight back to space (about 30%).
Greenhouse gas	A gas that absorbs and emits infrared radiation, such as carbon dioxide, methane, water vapor, nitrous oxide, and ozone.
Differential heating	Uneven heating of Earth because sunlight strikes the equator more directly than the poles.

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Term	Definition
Convection cell	A circular loop of air formed as warm air rises and cool air sinks.
Hadley, Ferrel, and polar cells	The three convection cells in each hemisphere, from equator to pole.
Coriolis Effect	An apparent force from Earth's rotation; deflects moving air to its right in the Northern Hemisphere.
Trade winds	Surface winds of the Hadley cell, deflected toward the west by the Coriolis Effect.
Temperature	An indirect measure of the average kinetic energy of a substance's randomly moving molecules.
Sulfate aerosols	A haze of tiny sulfate droplets formed from volcanic sulfur dioxide; reflects sunlight for a year or two.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
6	Main ingredients of Earth-based weather
5 mm to 15+ cm	Range of hailstone sizes (pea-sized to larger than a softball)
8 in (about 20 cm)	Largest hailstone on record in the United States
1 second	Of flash-to-thunder delay is about 1/5 mile, or about 1/3 kilometer
100%	Relative humidity at which cloud droplets form (air at its dew point)
10	Basic cloud types (genera) recognized by meteorologists
About 30%	Share of incoming sunlight Earth reflects straight back into space
3 or more	Atoms per molecule in nearly all greenhouse gases
About 60 °F	How much colder Earth's average surface would be without greenhouse gases
About 2.4 billion years ago	Earth likely froze over almost completely
3 per hemisphere	Convection cells: Hadley, Ferrel, and polar
32 °F and 212 °F	Freezing and boiling points of water at sea level, Fahrenheit
0 °C and 100 °C	Freezing and boiling points of water at sea level, Celsius
C = 5/9 (F − 32)	Fahrenheit-to-Celsius conversion
1991	Eruption of Mount Pinatubo, Philippines; cooled Earth about 0.5 C° (about 1 F°) the next year
1 to 2 years	How long volcanic sulfate aerosols linger in the stratosphere

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Value	What it refers to
About 1%	Volcanic CO_2 (a few hundred million metric tons per year) compared with human CO_2
150 years	Span of temperature records showing human-driven warming
1 to 3 ft (0.3 to 1 m)	Projected sea-level rise this century

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — What Weather Is

All Earth-based weather happens in the **troposphere**, so when we say "atmosphere" in the context of weather, we mean that lowest layer. From a physics standpoint, weather is a set of **thermodynamic events**: events driven by the movement of heat due to temperature differences. Weather is essentially the result of **energy transfer and energy transformation**.

Six main ingredients combine to create Earth-based weather: the atmosphere itself, water in all three phases, the Sun, Earth's rotation, volcanic activity, and biological activity.

Meteorology, the study of weather, comes from the Greek *meteoros*, "lofty" or "high in the air," so it literally means "the study of things high in the air." Without an atmosphere there is no weather. **The Moon** receives sunlight and swings through large temperature changes, but it lacks a substantial atmosphere to create the fluid dynamics weather needs.

Part 2 — Water: Precipitation, Hail, Lightning, and Thunder

Water is the **only common substance that naturally exists in all three phases** (ice, liquid water, and water vapor) within the temperatures and pressures of Earth's atmosphere, which is why it plays perhaps the most crucial role in weather.

Precipitation falls as liquid rain, as snow (fully formed ice crystal structures), or as **sleet**, small pellets of ice that form when falling raindrops freeze before they reach the ground. In violent storms with **cumulonimbus clouds**, strong updrafts carry ice particles up and down through supercooled water droplets that freeze on layer by layer, building **hailstones** from about **5 millimeters to more than 15 centimeters** across. The largest on record in the United States measured **8 inches**, about 20 centimeters.

Inside a thunderstorm cloud, collisions between ice crystals and **graupel**, soft hail pellets, create electric charge separations and enormous voltage differences, within the cloud and between the cloud and the ground. **Lightning** is the giant electrical discharge that dissipates that voltage difference, heating and ionizing the air along its path. It can strike within a cloud, cloud to cloud, cloud to ground, or even ground to cloud. Its energy breaks up nitrogen molecules, forming compounds that dissolve in rainwater as **nitrates** and help fertilize vegetation below.

Thunder is the sound of the shock wave made when lightning heats the air so suddenly that it expands explosively. Because sound travels much more slowly than light, each second between the flash and the thunder means about **one-fifth of a mile**, or about a third of a kilometer.

Part 3 — Clouds and Evaporation

Clouds form from very tiny water droplets or ice crystals that gather around **condensation nuclei**: tiny airborne specks of dust, smoke, sulfate particles, and bacteria. They form where the **relative humidity reaches 100%**, because the air has cooled to the **dew point** for air at that altitude.

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Wind, cold air fronts, or mountains lift and cool moist air. The droplets grow by colliding and merging, or by feeding the growth of ice crystals, until they are too heavy to stay aloft and precipitation begins. Clouds can appear throughout the **homosphere** (troposphere, stratosphere, and mesosphere), but almost all form in the troposphere. Meteorologists classify clouds into **ten basic types, called genera**, and **fog** is simply a cloud at ground level.

Water gets into the air by **evaporation**, the persistent transfer of fast-moving water molecules from liquid water to the air above it. The fastest molecules escape and take their kinetic energy with them, so **evaporation is a cooling process**. The rate depends on the puddle's size and depth, the warmth of the air and ground, wind, direct sunlight, and how low the surrounding relative humidity is.

When the vapor condenses into cloud droplets and ice crystals, it releases that energy as **latent heat**, warming the air. The warmer, more buoyant air fuels **updrafts** that keep the cloud aloft and growing.

† The episode says latent heat warms the air "directly below" the cloud's droplets. More precisely, the heat is released where condensation happens, warming the air inside the cloud and the rising air that feeds it. That extra buoyancy is what drives the updrafts.

Part 4 — Humidity and Dew Point

Relative humidity is the ratio of the current water vapor pressure to the saturation water vapor pressure, as a percentage: **RH = (current vapor pressure ÷ saturation vapor pressure) × 100%**. The **saturation vapor pressure** depends almost entirely on temperature; the warmer the air, the higher it is, so warmer air has more "elbow room" for water vapor.

If air below 100% relative humidity cools enough, its relative humidity reaches 100%; that temperature is the **dew point**. If the same air warms instead, its relative humidity falls.

That explains everyday effects. **Dew** forms overnight when grass, bushes, or a car cool to the dew point of the air around them. In **winter**, cold outside air holds very little water vapor even near 100% relative humidity, so when it is warmed indoors its relative humidity drops sharply. A **drought** is prolonged below-average precipitation in one location.

Part 5 — The Sun, Albedo, and Greenhouse Gases

Heating that drives weather comes almost exclusively from **the Sun**, a nearly constant, renewable source of radiant energy. Some of it warms the atmosphere directly; the rest warms lands and seas, which in turn warm the atmosphere.

About **30%** of incoming sunlight is reflected straight back into space, a fraction set by Earth's **albedo**, its reflectivity. How much of the rest Earth holds onto depends on **greenhouse gases**, gases that absorb and emit infrared radiation and send part of it back toward the surface. Nearly all have **three or more atoms per molecule**, so the two-atom nitrogen and oxygen that make up most of our air are not greenhouse gases. Prominent ones are **carbon dioxide, methane, water vapor, nitrous oxide, and ozone**.

Greenhouse gases keep Earth from freezing over: without them, the average surface temperature would be about **60 °F colder**. About **2.4 billion years ago**, rising oxygen likely destroyed much of the warming methane, and Earth froze over almost completely. Too much

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greenhouse gas would melt the **ice over land** and expand the warming seawater, raising sea levels dangerously.

† The episode says greenhouse gases absorb the infrared radiation Earth "reflects." Earth emits (radiates) infrared because it is warm. Reflection is the albedo part of the story: sunlight bounced straight back to space.

Part 6 — Differential Heating, Global Winds, and Temperature

Because Earth is roughly spherical, sunlight strikes the equator much more directly than the poles, and because Earth's axis is tilted, the pattern shifts with the seasons. Earth is **differentially heated**. Warm air rises and cool air sinks, so this sets up **convection cells**, circular loops of air, with **low pressure** below rising warm air and **high pressure** below sinking cool air.

Each hemisphere has three cells from equator to pole: the **Hadley**, **Ferrel**, and **polar** cells. The jet streams form near their boundaries, where sharp temperature contrasts combine with the **Coriolis Effect**, an apparent force caused by Earth's rotation. In the Northern Hemisphere it deflects moving air to the air's right; in the Hadley cell, high-altitude northbound winds shift east and southbound surface winds shift west, becoming the **trade winds**. The effect is negligible at the equator and grows toward the poles.

Temperature is an indirect measure of the average kinetic energy of randomly moving molecules. It is not the same as heat; in the very thin thermosphere, a high temperature would not make you feel warm. **Fahrenheit** puts water's freezing and boiling points at **32° and 212°**, **Celsius** at **0° and 100°**, and **C = 5/9 (F − 32)**.

Part 7 — Volcanoes, Human Activity, and Climate

Volcanoes mostly affect climate in the short term. An eruption can send fine ash and **sulfur dioxide** into the stratosphere. The ash falls out within weeks, but the sulfur dioxide forms **sulfate aerosols** that linger for a year or two, reflecting sunlight and cooling Earth. **Mount Pinatubo** (Philippines, **1991**) lowered the next year's global temperature by about half a Celsius degree, roughly one Fahrenheit degree. Volcanoes also release carbon dioxide, **a few hundred million metric tons per year**, but that is only about **1%** of what humans add by burning fossil fuels.

Over the last **150 years** of data, human activity has greatly increased Earth's average temperature. Burning fossil fuels, deforestation, and raising methane-producing animals have caused a sharp rise in greenhouse gases and a sustained long-term warming. Sea levels are projected to rise roughly **1 to 3 feet (a third of a meter to 1 meter)** this century, and potentially several meters over the coming centuries.

Climate change doesn't just mean warmer temperatures. Adding thermal energy **supercharges** the thermodynamic processes that drive weather, bringing more intense storms, more severe droughts, more extreme precipitation, and less predictable weather. Because the ingredients are interconnected, a change in one cascades through the whole system.

From the first raindrop to the global winds, every part of Earth's weather is physics at work: heat on the move, driven by the Sun and shaped by water, rotation, volcanoes, and life itself.

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TRY IT YOURSELF

Count the storm, convert the forecast. You see a lightning flash and count 15 seconds before you hear the thunder. About how far away was the strike, in miles and in kilometers? Later, the forecast high is 86 °F. Use C = 5/9 (F − 32) to convert it to Celsius.

(Answer: at about 1/5 mile or 1/3 kilometer per second, 15 seconds means about 3 miles, or about 5 kilometers. For the forecast, C = 5/9 × (86 − 32) = 5/9 × 54 = 30 °C. A warm summer day, and a storm close enough to head indoors.)

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — What Weather Is

1. In which layer of the atmosphere does all Earth-based weather occur? What do meteorologists usually mean by "atmosphere" when they talk about weather?

2. From a physics standpoint, what is a thermodynamic event, and why can weather be described as a set of them?

3. List the six main ingredients that combine to create Earth-based weather.

4. Where does the word meteorology come from, and what does it literally mean?

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5. The Moon receives sunlight and has large temperature changes. Why doesn't it have weather?

Part 2 — Water: Precipitation, Hail, Lightning, and Thunder

1. What makes water unique among the common substances in Earth's atmosphere?

2. Describe the difference between rain, snow, and sleet.

3. Describe how hailstones form. About how large can they get, and what is the U.S. record?

4. Explain how electric charge builds up in a thunderstorm cloud and what lightning is.

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5. How does lightning help fertilize vegetation?

6. What causes thunder? If you count 10 seconds between a lightning flash and its thunder, about how far away was the strike?

Part 3 — Clouds and Evaporation

1. What are condensation nuclei? Give three examples.

2. What is the immediate reason cloud droplets form at a particular altitude?

3. Explain why evaporation is a cooling process.

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4. List at least four factors that affect how fast a puddle evaporates.

5. What is latent heat, and how does its release help a cloud stay aloft and grow?

6. How many basic cloud types do meteorologists recognize, and what is fog?

Part 4 — Humidity and Dew Point

1. Define relative humidity and write the formula for it.

2. What does saturation vapor pressure depend on, and how?

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3. What is the dew point? What happens to the relative humidity of a parcel of air if it warms up?

4. Why do we often find dew on grass or cars in the early morning?

5. Why is indoor air often very dry in winter, even when it is snowing outside?

Part 5 — The Sun, Albedo, and Greenhouse Gases

1. What supplies nearly all the heating behind Earth's weather? Describe the two ways that energy reaches the atmosphere.

2. What is albedo, and about what fraction of incoming sunlight does Earth reflect straight back into space?

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3. What is a greenhouse gas? Why are nitrogen and oxygen not greenhouse gases? Name three greenhouse gases.

4. Why are greenhouse gases necessary? What likely happened about 2.4 billion years ago?

5. Give two reasons too much greenhouse gas would raise sea levels.

Part 6 — Differential Heating, Global Winds, and Temperature

1. Why is Earth heated unevenly, and why does that pattern change through the year?

2. How does differential heating create convection cells? Where do low- and high-pressure areas form?

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3. Name the three convection cells in each hemisphere, in order from the equator. Where do the jet streams form?

4. What is the Coriolis Effect? Which way does it deflect moving air in the Northern Hemisphere, and how does it produce the trade winds?

5. Define temperature. Why wouldn't the very high temperatures of the thermosphere make you feel warm?

6. Use C = 5/9 (F − 32) to convert 68 °F to Celsius. Show your work.

Part 7 — Volcanoes, Human Activity, and Climate

1. How can a large volcanic eruption cool Earth in the short term? Why doesn't the ash cause most of the cooling?

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2. What effect did the 1991 eruption of Mount Pinatubo have on global temperature?

3. How does the carbon dioxide released by all volcanic activity compare with the amount released by humans?

4. List three human activities that increase greenhouse gases. How much is sea level projected to rise this century?

5. Why does climate change mean more extreme weather, not just warmer temperatures?

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YOUR NOTES

Notes

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